Rounding Multi-Digit Numbers to Named Places
Answer Key
Grade 4–5 Mathematics

Quick Answers

1. 3000, 3500, 4000
2. 2700, 2650, 2600
3. 49000, 48500, 48000
4. 7900, 7950, 8000
5. 27000, 26500, 26000
6. 4100, 4150, 4200, 2800, 7000
7. 6000, 5500, 5000
8. 8000, 8500, 9000, 4000, 4315, —
9. 12500, 12000, 13000, —
10. 4600, 4600, —

Curriculum

Level: Grade 4–5 Mathematics
4.NBT.A.3 – *problems 1, 2, 3, 4, 5, 7, 9, 10*
4.NBT.B.4 – *problems 6, 8*
Difficulty 1–5 of 5 across 36 tagged checks.

Common wrong answers

5. *If they got 30000: rounded to the nearest ten thousand instead of the nearest thousand*

1. Round 3847 to the nearest thousand.
The thousands below and above are $\boxed{3000}$ and 4000, and halfway between them is $\boxed{3500}$.
3847 is past 3500, so it is nearer the upper benchmark.
$\boxed{4000}$

2. Round 2614 to the nearest hundred.
The hundreds around 2614 are 2600 and $\boxed{2700}$, and halfway is $\boxed{2650}$.
2614 has not reached 2650, so it stays with the lower hundred.
$\boxed{2600}$

Grading note: the tens digit 1 is what decides it. Looking at the ones digit 4, or at the whole "14", is the usual slip.

3. Round 48362 to the nearest thousand.
The thousands around it are 48000 and $\boxed{49000}$; halfway is $\boxed{48500}$.
48362 comes before 48500, so it rounds down. The digits to the right of the thousands place all become zeros, and the 48 thousands are untouched.
$\boxed{48000}$

4. Round 7952 to the nearest hundred.

The hundreds around it are $\boxed{7900}$ and 8000; halfway is $\boxed{7950}$.
7952 is 2 past 7950, so it rounds up — and rounding up here carries into the thousands place, turning 79 hundreds into 80 hundreds.
$\boxed{8000}$

Answer to "how do you know": the tens digit is 5 and the ones digit is 2, so the number sits just after the halfway mark.

5. Stadium crowd of 26481, to the nearest thousand.

The thousands around it are 26000 and $\boxed{27000}$; halfway is $\boxed{26500}$.
26481 is 19 short of 26500, so it has not reached halfway and rounds down.
$\boxed{26000}$

Grading note: 30000 means the crowd was rounded to the nearest ten thousand instead. Read which place the question names before rounding.

6. Estimate $4182 + 2759$ to the nearest hundred.

For 4182: the hundreds are $\boxed{4100}$ and 4200, halfway is $\boxed{4150}$, and 4182 is past it, so $\boxed{4200}$.
For 2759: halfway between 2700 and 2800 is 2750, and 2759 is past it, so $\boxed{2800}$.
Now add the rounded numbers: $4200 + 2800$.
$\boxed{7000}$

Check: the exact sum is 6941, so the estimate is close and easy to get mentally — which is the point of estimating.

7. Round 5499 to the nearest hundred, then to the nearest thousand.

Nearest hundred: the hundreds are 5400 and 5500, halfway is 5450, and 5499 is past it.
$\boxed{5500}$
Nearest thousand: the thousands are 5000 and $\boxed{6000}$, halfway is 5500, and 5499 is just *before* it.
$\boxed{5000}$

Answer to "why opposite directions": the two questions use different halfway marks. Against 5450 the number is above; against 5500 it is below, by one. Rounding a number that has already been rounded ($5500 \to 6000$) is exactly the mistake this problem is built to catch.

8. Estimate $8206 - 3891$ to the nearest thousand.

For 8206: the thousands are $\boxed{8000}$ and $\boxed{9000}$, halfway is $\boxed{8500}$, and 8206 is before it, so it rounds to 8000.
For 3891: halfway between 3000 and 4000 is 3500, and 3891 is past it, so it rounds to 4000.
Estimate: $8000 - 4000$.
$\boxed{4000}$
Exact: $8206 - 3891$.
$\boxed{4315}$
$\boxed{4000 < 4315}$, so the estimate is smaller than the exact difference.

Why: the first number was rounded down and the second was rounded up, and both of those changes make a difference smaller.

9. River trail 12438 m, ridge trail 12509 m.

Halfway between 12000 and 13000 is $\boxed{12500}$.
River: 12438 is before halfway, so $\boxed{12000}$.
Ridge: 12509 is past halfway, so $\boxed{13000}$.
The exact lengths compare as $\boxed{12438 < 12509}$, so the ridge trail is longer — and here the rounded numbers agree with that.

Worth saying to the student: they agree this time, but two numbers only 71 apart landed on rounded values 1000 apart. Rounded numbers are for reporting, not for deciding which of two close measurements is larger.

10. Challenge: which whole numbers round to 4600?

Rounding to the nearest hundred means choosing between 4500 and 4600 below, and between 4600 and 4700 above.
Least: halfway between 4500 and 4600 is 4550, and a number exactly on halfway rounds up, so 4550 is the smallest that reaches 4600: $\boxed{4600}$ is what it rounds to.
Greatest: halfway between 4600 and 4700 is 4650, which rounds up to 4700. So the largest whole number that still rounds down to 4600 is one less, 4649, and it rounds to $\boxed{4600}$.
In order: $\boxed{4550 < 4649}$

So every whole number from 4550 to 4649 rounds to 4600 — one hundred numbers in all.